



Motivation

Classical Galerkin schemes (finite elements, discontinuous Galerkin):

- Trial/test functions are piecewise polynomials;
- Discrete spaces applicable to a wide range of PDEs;
- Good accuracy may require large linear systems

Goal: To design **high-order** numerical methods with **reduced computational cost**, using discrete spaces that are tailored to the PDE itself.

Fields: Wave problems in acoustics, electromagnetics, etc.

Trefftz and Quasi-Trefftz Methods

Trefftz methods. High-order discontinuous Galerkin schemes whose trial and test functions are **exact** solutions of the PDE on every mesh element $T \in \mathcal{T}_h$.

Example: Plane waves $e^{i\omega \mathbf{d} \cdot \mathbf{x}}$ for Helmholtz or Maxwell's equations.

✓ Accuracy with reduced computational cost for linear, homogeneous PDEs with piecewise-*constant* coefficients

✗ *Not* applicable for *variable* coefficients (e.g., wave propagation in inhomogeneous media).

⇒ **Quasi-Trefftz methods.** Trial/test functions are only element-wise **approximate** solutions, preserving the favorable properties of Trefftz schemes while gaining flexibility for **variable coefficients**.

Quasi-Trefftz Spaces: Scalar Problems

$$\mathcal{L}u = 0 \quad \text{in } \Omega \subset \mathbb{R}^d$$

where $\mathcal{L}$ is a linear differential operator of order γ with smooth coefficients.

Local quasi-Trefftz space on element T , centered at $\mathbf{x}_T \in T$:

$$\mathbb{Q}\mathbb{T}_p(T) = \{ Q \in \mathbb{P}_p : \mathcal{T}_{p-\gamma}^{\mathbf{x}_T} \circ \mathcal{L} Q = 0 \} \subset \mathbb{P}_p$$

$\mathcal{T}_q^{\mathbf{x}_T}$: Taylor operator of order q at $\mathbf{x}_T$

Functions in $\mathbb{Q}\mathbb{T}_p(T)$ solve the PDE up to a residual $O(\|\mathbf{x} - \mathbf{x}_T\|^{p-\gamma+1})$.

Main results (scalar case):

- Efficient *layered* construction algorithm, from the decomposition $\mathbb{P}_p = \bigoplus_{k=0}^p \tilde{\mathbb{P}}_k$ and surjectivity of the highest-order part of $\mathcal{L}$ with frozen coefficients;
- Explicit bases, dimension $O(p^{d-1})$ instead of $O(p^d)$;
- Optimal-order **best-approximation** estimates.

Extension to Time-Harmonic Maxwell's Equations

Fields $\mathbf{E}, \mathbf{B}$, wavenumber $\kappa > 0$, variable scalar coefficients $\varepsilon(\mathbf{x}), \mu(\mathbf{x})$:

$$\begin{cases} \nabla \times \mathbf{E} - i\kappa\mu\mathbf{B} = \mathbf{0} & \text{in } \Omega \\ \nabla \times \mathbf{B} + i\kappa\varepsilon\mathbf{E} = \mathbf{0} & \text{in } \Omega \end{cases}$$

Quasi-Trefftz space for the Maxwell system, on T centered at $\mathbf{x}_T$:

$$\mathbb{Q}\mathbb{T}_p(T) := \left\{ (Q^E, Q^B) \in (\mathbb{P}_p)^3 \times (\mathbb{P}_p)^3 : \begin{cases} \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \times Q^E - i\kappa\mu Q^B) = \mathbf{0} \\ \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \times Q^B + i\kappa\varepsilon Q^E) = \mathbf{0} \\ \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \cdot (\mu Q^B)) = 0 \\ \mathcal{T}_{p-1}^{\mathbf{x}_T}(\nabla \cdot (\varepsilon Q^E)) = 0 \end{cases} \right\}$$

Helmholtz-Type Decomposition

Key difficulty: Unlike the scalar case, the curl operator is *not* surjective on homogeneous polynomial vector fields $\tilde{\mathbb{P}}_k^3$, it is only surjective onto **divergence-free** fields:

$$\text{Range}(\nabla \times |_{(\tilde{\mathbb{P}}_{k+2}^3)}) = \ker(\nabla \cdot |_{(\tilde{\mathbb{P}}_{k+1}^3)})$$

Helmholtz-type decomposition. Every homogeneous polynomial vector field splits as

$$(\tilde{\mathbb{P}}_k^3) = \underbrace{\tilde{\mathbb{H}}_k}_{\text{harmonic}} \oplus \underbrace{\tilde{\mathbb{S}}_k^*}_{\text{solenoidal}} \oplus \underbrace{\tilde{\mathbb{I}}_k^*}_{\text{irrotational}}$$

$$\tilde{\mathbb{S}}_k = \ker(\nabla \cdot |_{(\tilde{\mathbb{P}}_{k+1}^3)}), \tilde{\mathbb{I}}_k = \ker(\nabla \times |_{(\tilde{\mathbb{P}}_{k+2}^3)}), \tilde{\mathbb{H}}_k = \tilde{\mathbb{S}}_k \cap \tilde{\mathbb{I}}_k, \tilde{\mathbb{S}}_k = \tilde{\mathbb{S}}_k^* \oplus \tilde{\mathbb{H}}_k, \tilde{\mathbb{I}}_k = \tilde{\mathbb{I}}_k^* \oplus \tilde{\mathbb{H}}_k.$$

A quasi-Trefftz function is built **layer by layer**, degree $k = 1, \dots, p$, solving small linear problems for the parts solenoidal $\mathbf{F}_k^*$, irrotational $\mathbf{G}_k^*$, given harmonic components $\mathbf{H}_k^*$:

$$\mathbf{Q}^* = \mathbf{H}_0^* + \sum_{k=1}^p (\mathbf{H}_k^* + \mathbf{F}_k^* + \mathbf{G}_k^*) \quad \star = E, B$$

Layered Construction Algorithm

(Input) $p \in \mathbb{N}$, $\mathbf{x}_T \in T$, $\varepsilon_k, \mu_k \in \tilde{\mathbb{P}}_k$, $\mathbf{H}_k^E, \mathbf{H}_k^B \in \tilde{\mathbb{H}}_k$, $k = 0, \dots, p$

(Step $k = 1, \dots, p$)

▷ Find $\mathbf{F}_k^E, \mathbf{F}_k^B \in \tilde{\mathbb{S}}_k^*$ such that

$$\begin{cases} \nabla \times \mathbf{F}_k^E = i\kappa \left(\mu_{k-1} \mathbf{H}_0^B + \sum_{j=1}^{k-1} \mu_{k-1-j} (\mathbf{F}_j^B + \mathbf{G}_j^B + \mathbf{H}_j^B) \right) \\ \nabla \times \mathbf{F}_k^B = -i\kappa \left(\varepsilon_{k-1} \mathbf{H}_0^E + \sum_{j=1}^{k-1} \varepsilon_{k-1-j} (\mathbf{F}_j^E + \mathbf{G}_j^E + \mathbf{H}_j^E) \right) \end{cases}$$

▷ Find $\mathbf{G}_k^E, \mathbf{G}_k^B \in \tilde{\mathbb{I}}_k^*$ such that

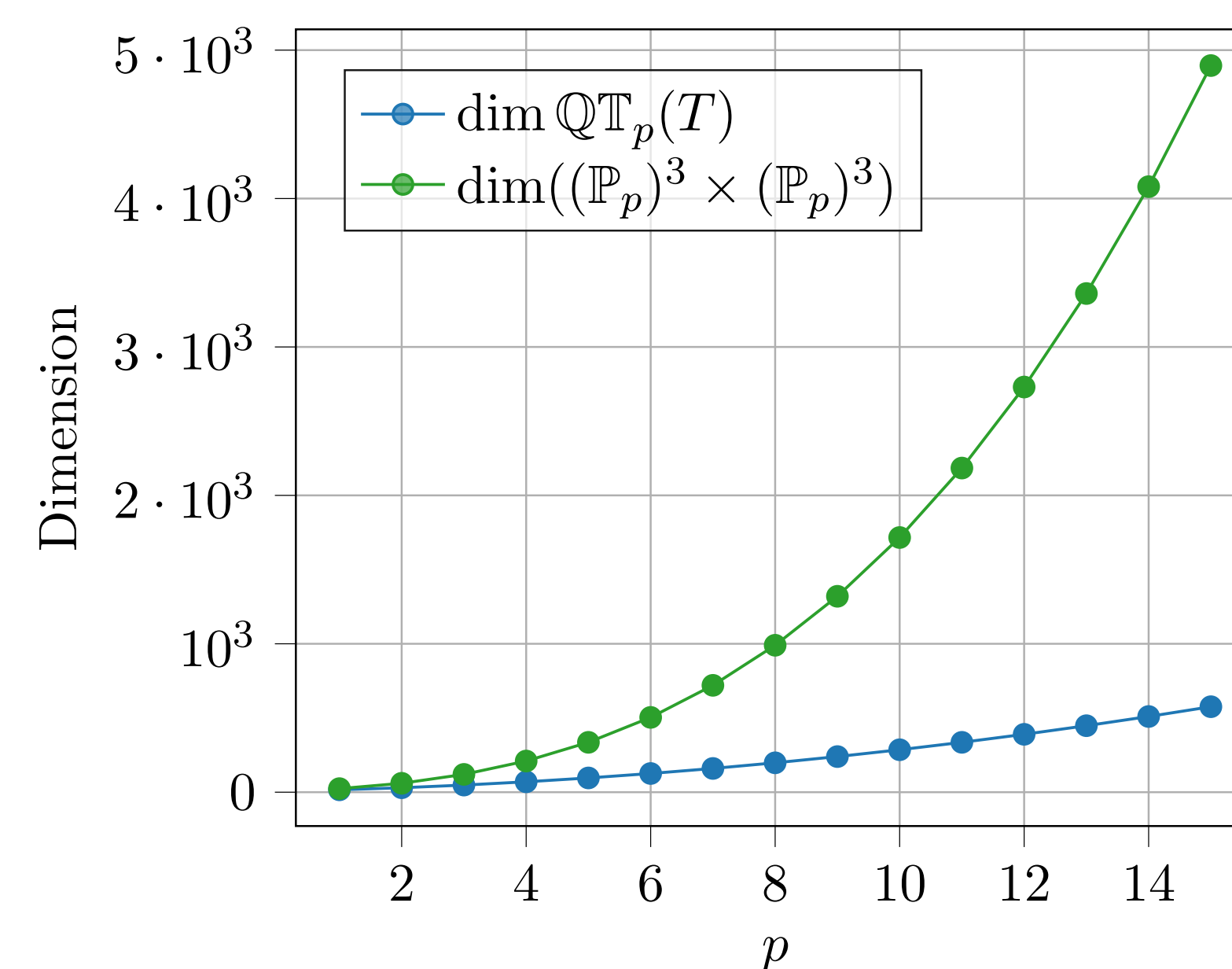
$$\begin{cases} \nabla \cdot \mathbf{G}_k^E = -\varepsilon_0^{-1} \nabla \cdot \left(\varepsilon_k \mathbf{H}_0^E + \sum_{j=1}^{k-1} \varepsilon_{k-j} (\mathbf{F}_j^E + \mathbf{G}_j^E + \mathbf{H}_j^E) \right) \\ \nabla \cdot \mathbf{G}_k^B = -\mu_0^{-1} \nabla \cdot \left(\mu_k \mathbf{H}_0^B + \sum_{j=1}^{k-1} \mu_{k-j} (\mathbf{F}_j^B + \mathbf{G}_j^B + \mathbf{H}_j^B) \right) \end{cases}$$

(Output) $(Q^E, Q^B) \in \mathbb{Q}\mathbb{T}_p(T)$; $\mathbf{Q}^* = \mathbf{H}_0^* + \sum_{k=1}^p (\mathbf{H}_k^* + \mathbf{F}_k^* + \mathbf{G}_k^*)$, $\star = E, B$

Basis and Dimension of the Quasi-Trefftz Space

$$\dim \mathbb{Q}\mathbb{T}_p(T) = \dim(\mathbb{H}_p \times \mathbb{H}_p) = 2(p+1)(p+3) = O(p^2)$$

$$\dim((\mathbb{P}_p)^3 \times (\mathbb{P}_p)^3) = (p+1)(p+2)(p+3) = O(p^3)$$



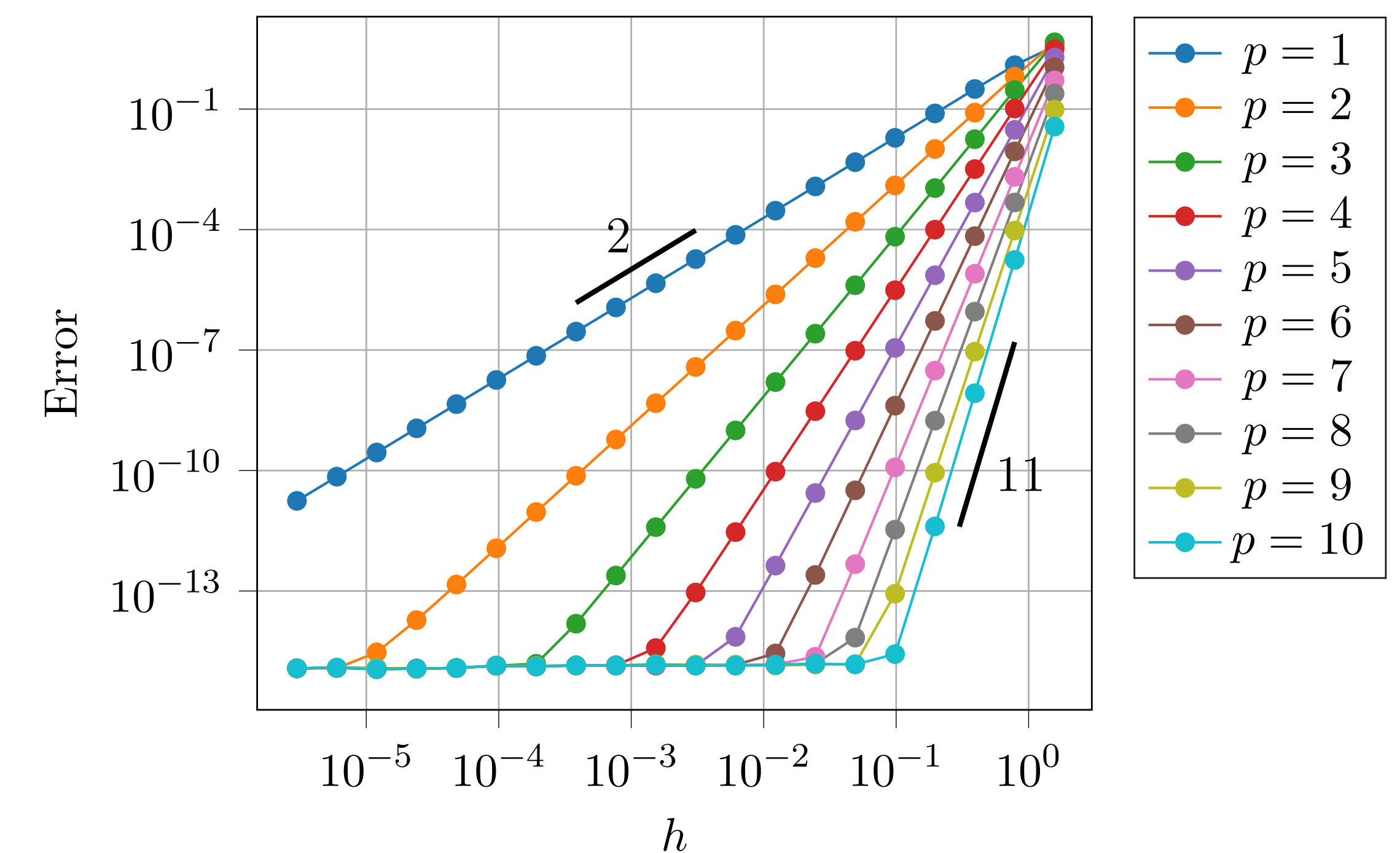
Approximation Properties & Numerical Validation

Theorem (Best Approximation Properties). Given $p \in \mathbb{N}$ and $T \subset \mathbb{R}^3$, suppose $\varepsilon, \mu \in C^p(T; \mathbb{C})$ and $(\mathbf{E}, \mathbf{B}) \in (C^{p+1}(T; \mathbb{C}^3))^2$ solves Maxwell's equations in T . Then,

$$(\mathcal{T}_p^{\mathbf{x}_T} \mathbf{E}, \mathcal{T}_p^{\mathbf{x}_T} \mathbf{B}) \in \mathbb{Q}\mathbb{T}_p(T),$$

and for T star-shaped w.r.t. $\mathbf{x}_T$,

$$\inf_{(Q^E, Q^B) \in \mathbb{Q}\mathbb{T}_p(T)} (\|\mathbf{E} - Q^E\| + \|\mathbf{B} - Q^B\|) \leq \frac{3^{p+1}}{(p+1)!} h_T^{p+1} (\|\mathbf{E}\| + \|\mathbf{B}\|).$$



Error: $\sup_{\|\mathbf{x} - \mathbf{x}_T\| \leq h} (\|\mathbf{E}(\mathbf{x}) - Q^E(\mathbf{x})\| + \|\mathbf{B}(\mathbf{x}) - Q^B(\mathbf{x})\|)$

Conclusions & Perspectives

- Quasi-Trefftz spaces retain the approximation properties of Trefftz methods while accommodating **variable coefficients**, with dimension *smaller* than the full polynomial space.
- The layered construction, developed for scalar PDEs, extend to the **first-order time-harmonic Maxwell system** with some extra tools like the Helmholtz-type decomposition.

Ongoing and future work:

- Quasi-Trefftz **DG variational formulation** for Maxwell's equations, and its implementation
- Extension to further PDEs
- Development of *a posteriori* error estimation and adaptive algorithms.

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